

SHAHEER FARDAN

Complete AI / ML Project Portfolio

LLM Systems • Model Optimization • Production MLOps • Computer Vision • Medical AI • Generative AI • NLP

20+

benchmarked projects

10–11×

LLM throughput

73.7%

model reduction

0.968

thesis F1

Architecture → Experimentation → Optimization → Serving → Deployment → Production Engineering

Portfolio Map

LLM Inference & Compression

6 projects

Production MLOps

3-project suite

LLM Fine-Tuning & Alignment

6+ model/application tracks

NLP & Transformers

classification • translation • routing

CV Detection & Segmentation

5 applied systems

Medical Imaging

4 segmentation/classification tracks

Classification & Representation

5 projects

Generative / Multimodal

5 projects

Graduate Thesis Research

3 strategies • 400+ experiments

Plus: 15+ years of enterprise systems engineering underpinning production-first AI reasoning.

PROJECT DOMAIN

LLM Inference, Optimization & Compression

Systematic experiments across latency, throughput, memory, model size and quality.

Llama-3.2 Serving Pipeline

24

experiments

10–11×

throughput

92%

lower per-sample latency

ROUGE-1

quality tracked

- News-headline generation with Llama-3.2-1B; benchmarked latency, throughput, peak memory and ROUGE-1.
- Layered KV caching, structured pruning, quantization, speculative decoding (1B draft / 3B target), DeepSpeed tensor/pipeline parallelism and batch inference.
- KV caching + batching produced the standout performance combination.

Optimization logic

Change one execution variable at a time, profile the system, and retain generation quality as a hard constraint.

Systems signal

Covers autoregressive decode reuse, distributed inference overheads, batching trade-offs and model compression.

Compression & Hardware-Aware Optimization

MobileNetV3 Mobile Compression

5 techniques + 5 pipelines. QAT + graph optimization: 73.7% smaller, 6.9× faster CPU inference, -0.6% accuracy; TorchScript Mobile.

Compressed RoBERTa — Fake News

99.94% baseline test accuracy; pruned up to 85% of attention heads, up to 30% faster CPU inference and ~69% smaller model.

YOLOv8 Size-Specialized Architecture

Modified YAML to remove detection heads irrelevant to target object-size regime; improved inference speed with marginal accuracy reduction.

Clinical ResNet-18 Optimization

Depthwise-separable convs + inverted residuals + ONNX optimization across 16 experiments; ~96% smaller, ~98% lower memory, ~68% lower latency, 98.97% sensitivity.

GPT-2 Optimization & Deployment

Attention-head pruning, targeted fine-tuning and INT8 PTQ for resource-constrained CPU inference; characterized size/latency/quality trade-offs.

PROJECT DOMAIN

Production MLOps

A repeatable SageMaker production pattern generalized from CNNs to Vision Transformers to NLP.



AWS SageMaker — 3-Project Production Suite

Inventory Counting — ResNet-50 / EfficientNet-B0

Custom Docker/ECR, HPO, multi-instance training, model selection and full production lifecycle.

Dog-Breed Classification — ViT

Replicated the deployment pattern with a Vision Transformer to validate architectural portability.

Sentiment Classification — BERT

Extended the same production template to NLP, demonstrating cross-domain repeatability.

Production lifecycle

Train

Evaluate

Register

Deploy

A/B Test

Autoscale

Failover

API

Multi-model endpoints • SageMaker Pipelines • Lambda • API Gateway • Gradio • simulated failover validation

PROJECT DOMAIN

LLM Fine-Tuning, Alignment & Applications

PEFT, post-training, reasoning, retrieval and from-scratch sequence modeling.

LLM Fine-Tuning Suite

OPT-350M / BLOOMZ-560M

Stylistically consistent quote generation

OPT-350M + CodeAlpaca

Code generation on 10K samples

Phi-2

Product name / description generation

Phi-2 + Guanaco

Chat-format instruction tuning

BART + SAMSum

Dialogue summarization

GPT-2 + Reuters

Domain-styled text generation

Methods: LoRA • QLoRA • soft-prompt tuning • SageMaker training

Alignment, Reasoning & Reward Modeling

Reward Model

RoBERTa-base reward model for RLHF-style downstream pipelines.

DPO — Helpfulness

Qwen2.5-1.5B-Instruct aligned with QLoRA + Direct Preference Optimization.

DPO — Identity Conditioning

Controlled identity-conditioning experiment on Qwen2.5-1.5B-Instruct.

GRPO — GSM8K

Llama-3.2-1B-Instruct fine-tuned on 2,000 GSM8K samples with custom rewards and structured reasoning tags.

Instruction Tuning Study

Instruction fine-tuning evaluated with BLEU / ROUGE.

Translation, RAG & Intelligent Routing

Transformer Machine Translation — From Scratch

Built complete encoder + decoder blocks for bidirectional English–French translation on 46,798 pairs. Custom PyTorch training/data pipelines, dedicated tokenizers, BLEU/BERTScore evaluation and Gradio deployment.

Retrieval-Augmented Generation

Built RAG first from fundamentals using embeddings + cosine retrieval + grounded generation, then a LangChain + Chroma real-estate recommendation assistant with personalized listing rewriting.

AI-Powered Intent Classification & Request Routing

Fine-tuned DistilBERT on Databricks Dolly 15K to classify five request types and act as the routing core for a multi-model AI system, separating intent understanding from downstream execution.

PROJECT DOMAIN

Transformer & NLP Systems

From-scratch architectures plus production-oriented fine-tuning and controlled PEFT benchmarking.

Transformer & Text Classification Projects

Encoder vs Decoder — Sentiment

Custom PyTorch transformers from scratch: encoder 83.63% vs decoder 81.08% test accuracy.

Named Entity Recognition

DistilBERT fine-tuning: 98.4% validation accuracy.

Spam Detection

DistilBERT fine-tuning: 99% test accuracy.

Hate Speech Classification

ALBERT-v2 fine-tuning: 93.65% test accuracy.

PEFT Benchmarking

10 experiments on GPT-2 / RoBERTa-base varying LoRA rank and quantization precision; tracked F1, efficiency and adapter size.

PROJECT DOMAIN

Computer Vision — Detection & Segmentation

Applied vision systems spanning detection, OCR, fine-structure segmentation and scene understanding.

Detection, OCR & Applied Vision

Automated License Plate Recognition

Two-stage detect-then-read system: fine-tuned YOLOv26 for plate localization + fine-tuned TrOCR for character recognition; applicable to tolling, parking and fleet tracking.

Traffic-Light State Recognition

Five-class image/video detection. Benchmarked RT-DETR, YOLOv8 and YOLOv12; YOLOv8x @ 768 px achieved 52.6% mAP(50-75) and 86% mAP(50).

Retinal Vessel Segmentation

Benchmarked SegFormer, ResUNet and CBAM-ResUNet under severe imbalance; standard ResUNet reached 77.2 mIoU.

Multiclass Road Scene Segmentation

9 classes / 700 images. DeepLabV3 from scratch vs CBAM-ResUNet-50; DeepLabV3 won with 0.7828 validation mIoU vs 0.7220.

Human-Centric Vision Suite

Shared backbone: CBAM-Attention ResUNet-50 • validated on video input

Face Segmentation

Identity-preserving automated privacy blurring.

Hair Segmentation

Pixel-level virtual hair recoloring.

Full-Body Segmentation

Background removal for e-commerce / virtual studio.

Portrait Background Blur

Portrait-mode effect without depth-sensing hardware.

PROJECT DOMAIN

Medical Imaging

Segmentation under extreme imbalance plus comparative classification across X-ray and CT modalities.

Medical Segmentation Projects

0.8702

brain tumor mIoU

0.8028

cell nuclei mIoU

77.2%

retinal vessel mIoU

Brain Tumor Segmentation

3,000 scans + masks. CBAM-ResUNet-50 trained with Dice Loss + Categorical Cross-Entropy to address severe pixel-level class imbalance.

Cell Nuclei Segmentation

670 microscopy images. Reused the CBAM-ResUNet-50 architecture and combined Dice + CCE objective for another fine-structure segmentation task.

Architecture reuse across retinal vessels, tumors and nuclei demonstrates a consistent experimental segmentation framework.

COVID-19 Medical Image Classification

Chest X-Ray

Converted raw images to DICOM. Trained ResNet-18, ResNet-50 and ViT from scratch; fine-tuned EfficientNetB0, VGG16 and ViT. Best result: EfficientNetV2B0 at 98.22%.

Chest CT

Converted CT scans to DICOM. Trained VGG16 from scratch plus fine-tuned EfficientNetB0, VGG16, ViT and ResNet-50. From-scratch VGG16 reached 97.5%, outperforming best fine-tuned configuration (85%).

Key experimental finding: architecture / transfer-learning advantage was modality- and dataset-dependent rather than universal.

PROJECT DOMAIN

Image Classification & Representation Learning

Architecture benchmarking, transfer learning, low-data learning, ViT implementation and explainability.

Classification & Representation Projects

Museum Painting Genre

ResNet-18 scratch: 80%. Benchmarked AlexNet, ResNet-50, VGG-19, EfficientNet-B0/B4 and ViT; ViT led at 92%.

Landmark Recognition

Fine-tuned ResNet-50: 77% vs from-scratch CNN: 61%.

MNIST Digit Recognition

From-scratch CNN reached 99.43%.

Low-Data Classification

VGG16 transfer learning with <1,000 images; >90% validation accuracy within four epochs.

Vision Transformer — Scratch

Implemented patch embedding, positional encoding and multi-head self-attention from first principles on CIFAR-10.

Grad-CAM Explainability

Validated that predictions were driven by semantically meaningful features rather than dataset artifacts.

PROJECT DOMAIN

Generative AI & Vision-Language Models

Adversarial generation, diffusion and cross-modal representation learning.

Generative & Multimodal Projects

DCGAN — CelebA

Human-face generation; adversarial training dynamics and latent-space stabilization.

Conditional GAN — MNIST

Conditional image generation with quantitative evaluation.

DDPM — MNIST

Diffusion generation compared with cGAN; evaluated via FID and downstream CNN recognition.

Diffusion Study

Further diffusion-model experimentation through NVIDIA's Generative AI with Diffusion Models program.

CLIP

Built and trained a CLIP model for aligned visual-text representations.

Image Captioning

Built and trained an automatic image-captioning model for visual-semantic alignment.

PROJECT DOMAIN

Graduate Thesis Research

Generative AI + anomaly detection + classification for extreme class imbalance.

Fraud Detection Under Extreme Class Imbalance

284,807

transactions

492

fraud cases

0.17%

minority class

48

papers reviewed

Strategy 1 — Class Balancing

SMOTE • ADASYN • WGAN-GP • VAE across Logistic Regression, Decision Tree, Random Forest, XGBoost CPU/GPU and DNN.

Strategy 2 — Anomaly Detection

Standalone + novel two-level composite Autoencoder/VAE models; RMSE, Wasserstein Distance and MAE threshold studies.

Strategy 3 — CBABCAD

Original architecture combining classification and composite anomaly-detection subsystems, designed to be balancing-method / classifier agnostic.

Thesis — Experimental Scale & Results

323

WGAN-GP configs

24

VAE configs

96

threshold experiments

0.968

best CBABCAD F1

Generative oversampling

WGAN-GP was strongest oversampling method; best F1 0.905 with tuned XGBoost. SMOTE/ADASYN increased recall but at high false-positive cost.

Composite anomaly detection

Reduced false positives by 50%+ versus standalone anomaly-detection models.

CBABCAD result

0.947 recall • 0.989 precision • 0.968 F1. Even an unbalanced dataset fed directly to the composite anomaly subsystem achieved 0.951 F1, indicating a strong automation / standardization advantage.

What the Complete Portfolio Demonstrates

MODEL

CNN • Transformer
GAN/VAE • ViT

TRAIN

Scratch • transfer
PEFT • alignment

OPTIMIZE

QAT • pruning
distill • graph

SERVE

KV cache • batch
speculate • TP/PP

DEPLOY

ONNX • TorchScript
SageMaker • APIs

VALIDATE

Latency • memory
quality • SLA

Breadth is not the message. The message is continuity across the AI lifecycle.

The projects connect model architecture, experimentation, post-training, inference optimization, deployment and production constraints — with measurable trade-offs rather than isolated accuracy scores.

DEEP LEARNING • GENERATIVE AI • AI SYSTEMS • PRODUCTION ENGINEERING

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